UNIT 1.INTRODUCTION TO DATA WAREHOUSE

Data Warehouse Architecture

Data Warehouse Architecture refers to the **overall design and structure** of a data warehouse system.

It explains **how data is collected from multiple sources, processed, stored, and presented** to users for analysis and decision making.

A data warehouse stores **historical and integrated data** from different sources. The architecture may vary based on the **needs and size of an organization**.

Common Types of Data Warehouse Architecture

1. **Basic Data Warehouse Architecture**
   - Data is loaded directly from sources into the data warehouse.
2. **Data Warehouse Architecture with Staging Area**
   - A staging area is used to clean and transform data before storing it.
3. **Data Warehouse Architecture with Staging Area and Data Marts**
   - Includes data marts for specific departments like Sales or HR.

Characteristics of Data Warehouse Architecture

1. Scalability

- Can handle **large and growing volumes of data**
- Supports more users without performance issues

2. Extensibility

- New data sources and tools can be added easily
- No major changes required to existing structure

3. Security

- Protects data using access control and encryption
- Prevents unauthorized access

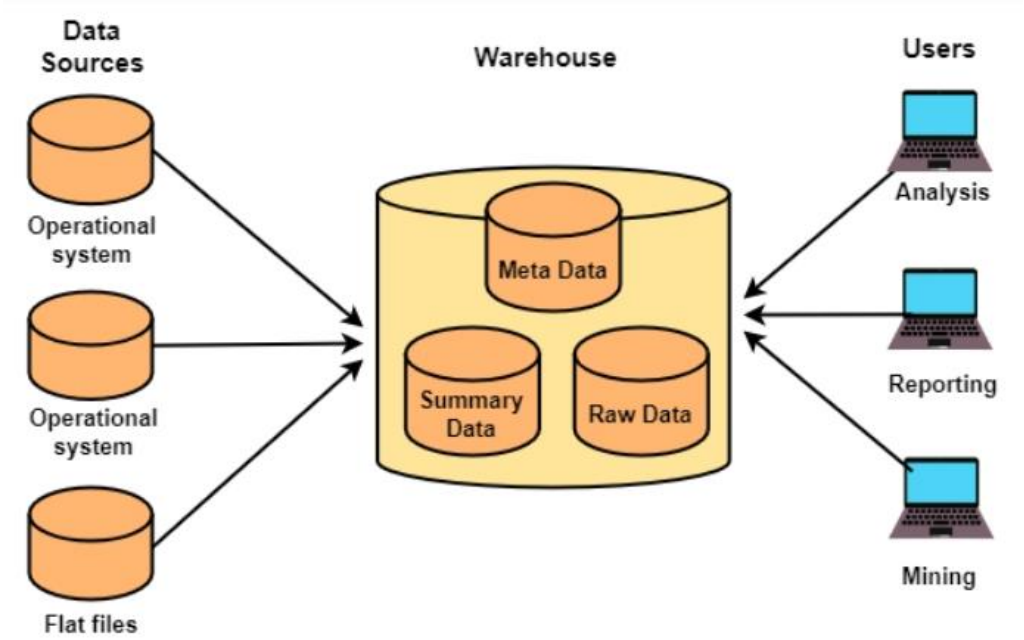
4. Administerability

- Easy to manage and monitor
- Supports performance tracking and maintenance

5. Analytical Processing

- Designed for **analysis and reporting**
- Supports decision making, not daily transactions

Basic Data Warehouse Architecture



1. **Operational System:** In data warehousing, an operational system is a system that processes an organization's day-to-day transactions.

2. **Flat File System:** A flat file system is a collection of files in which each file must have a unique name.

3. **Metadata:** The metadata contains information about other data but not the data itself, such as a message's text or an image's content. We use metadata to direct a query to the most relevant data source.

4. **Raw data:** Raw data is a set of data that has not yet been processed by machine or human.

5. **Summary data:** is short and clear information created from a large amount of data. It shows the main results or conclusions without detailed data. In data mining, summary data helps users quickly understand important information and make decisions.

Components of Data Warehouse Architecture

A data warehouse architecture consists of several key components that work together to store, manage, and analyze large volumes of data. The main components are:

- Data Sources – Various structured and unstructured data sources like databases, APIs, and flat files.

- ETL (Extract, Transform, Load) Process – Extracts data from sources, transforms it into a usable format, and loads it into the data warehouse.

- Data Staging Area – A temporary storage space where raw data is cleaned and transformed before loading.

- Data Warehouse Storage – The central repository where processed data is stored, often structured in schemas like star or snowflake.

- Metadata Repository – Stores information about data definitions, structures, and relationships to support data management.

- Data Marts – Subsets of the data warehouse designed for specific business functions or departments.

- OLAP (Online Analytical Processing) Engine – Enables complex queries, aggregations, and multi-dimensional data analysis.

- BI (Business Intelligence) and Reporting Tools – Used for data visualization, dashboards, and generating reports to support decision-making.

- Data Governance and Security – Ensures data quality, access control, compliance, and protection against unauthorized access.

Three-Tier Data Warehouse Architecture

The **Three-Tier Data Warehouse Architecture** is a widely used architecture that organizes the data warehouse system into three layers: **Bottom Tier, Middle Tier, and Top Tier**. Each tier has a specific role in storing, processing, and presenting data for decision-making.

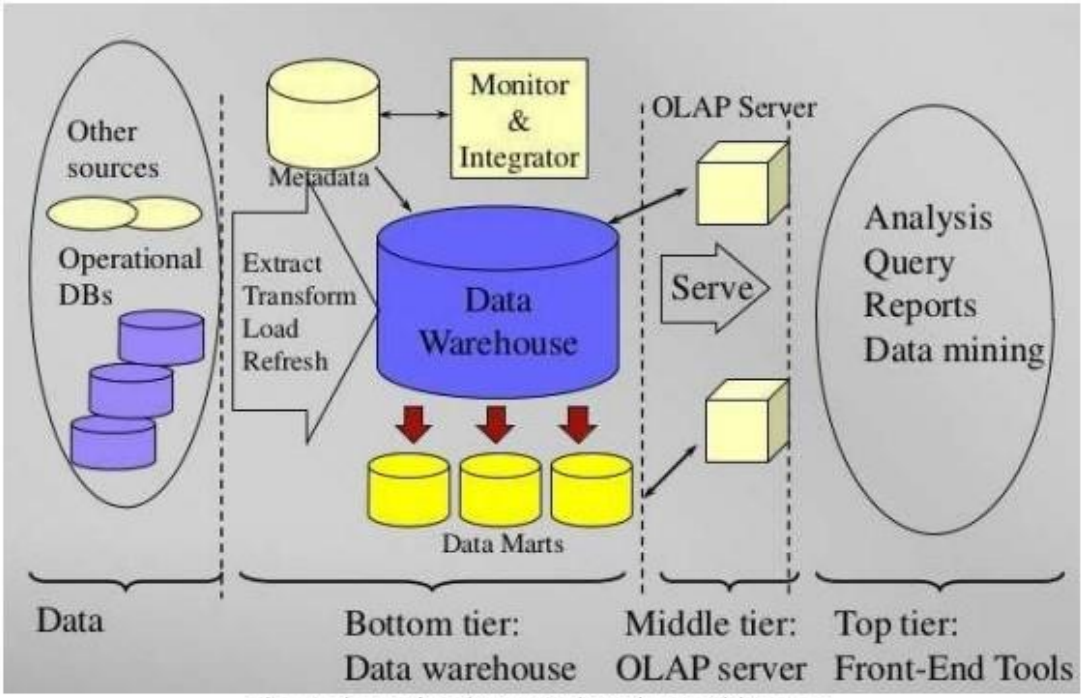


Fig: A three-tier data warehousing architecture.

1. Bottom Tier (Data Source and Data Storage Layer)

The bottom tier is the **base layer** of the data warehouse architecture. It includes various **data sources** such as operational databases, transaction systems, ERP, CRM systems, flat files, and external data sources.

Data from these sources is collected using **ETL (Extract, Transform, Load)** tools.

- **Extract**: Data is taken from different source systems.
- **Transform**: Data is cleaned by removing errors, duplicates, and inconsistencies. It is also integrated and formatted.
- **Load**: The transformed data is stored in the data warehouse.

This tier stores **large amounts of historical and summarized data** in a structured form using schemas like star or snowflake schema.

Example:

A retail company collects daily sales data from all branch billing systems. This data is cleaned and stored in a central data warehouse for long-term analysis.

2. Middle Tier (OLAP Server / Processing Layer)

The middle tier is the **analysis and processing layer**. It uses **OLAP (Online Analytical Processing)** servers to process data stored in the warehouse.

This tier supports analytical operations such as:

- Aggregation (total sales, average profit)
- Drill-down (year → month → day)
- Roll-up (day → month → year)
- Slicing and dicing of data

Types of OLAP servers include:

- **ROLAP** – uses relational databases
- **MOLAP** – uses multidimensional data cubes
- **HOLAP** – combination of both

The middle tier improves query performance and enables fast analysis.

Example:

A manager wants to know monthly sales of a product category in a particular region. The OLAP server quickly processes this request and provides results.

3. Top Tier (Presentation / User Interface Layer)

The top tier is the **front-end layer** used by end users. It provides tools such as reports, dashboards, charts, and data visualization tools.

Users such as managers, analysts, and executives use this tier to:

- View summarized reports
- Analyze trends and patterns
- Perform ad-hoc queries

This tier presents data in a **user-friendly and understandable format**.

Example:

A dashboard showing total yearly revenue, best-selling products, and regional performance helps top management in decision-making.

Advantages of Three-Tier Architecture

- Better performance and faster query processing
- Clear separation of storage, processing, and presentation
- Easy maintenance and scalability

- Supports effective decision-making

Difference between OLTP and OLAP

Introduction

Modern organizations use database systems for both **daily operations** and **decision making**.

To support these needs, two different processing systems are used:

- **OLTP (Online Transaction Processing)**
- **OLAP (Online Analytical Processing)**

Although both use databases, their **purpose, data structure, and usage** are very different.

OLTP (Online Transaction Processing)

OLTP systems are designed to handle **day-to-day business transactions**. They focus on **speed, accuracy, and consistency** of data.

Characteristics of OLTP

- Handles a **large number of small transactions**
- Transactions include **insert, update, and delete**
- Data is **current and detailed**
- Supports **many concurrent users**
- Uses **highly normalized tables** to reduce redundancy
- Requires **fast response time**

Example

- Banking system (deposit, withdrawal)
- Railway reservation system
- Online shopping order processing

OLAP (Online Analytical Processing)

OLAP systems are designed for **data analysis and decision support**. They use historical data and perform **complex queries** for business insights.

Characteristics of OLAP

- Handles **complex analytical queries**
- Data is **historical and summarized**
- Used mainly for **reading and analysis**
- Supports **fewer users**
- Uses **denormalized schemas** like star or snowflake
- Response time is **not as critical** as OLTP

Example

- Sales trend analysis
- Profit and revenue reporting
- Forecasting and business intelligence

Detailed Comparison Between OLTP and OLAP

Aspect	OLTP	OLAP
Full Form	Online Transaction Processing	Online Analytical Processing
Main Purpose	Daily transaction processing	Data analysis and decision support
Data Type	Current, detailed data	Historical, summarized data
Operations	Insert, update, delete	Read, aggregate, analyze
Query Complexity	Simple queries	Complex queries
Transactions	Many short transactions	Few long queries
Users	Clerks, customers, operators	Managers, analysts
Database Size	Small to medium	Very large
Data Updates	Frequent	Rare (mostly read-only)
Schema Design	Highly normalized	Denormalized (Star/Snowflake)
Response Time	Very fast	Slower than OLTP

Aspect	OLTP	OLAP
Data Integrity	Very high	Moderate
Example Systems	ATM, billing system	Data warehouse, BI tools

Multidimensional Data Model

A Multidimensional Data Model (MDM) organizes data into multiple dimensions such as time, product, location to support fast analytical queries in data warehouses and OLAP systems.

One Dimension

Sales of Various Products

Jam	Bread	Sugar	Milk
350	380	60	85

Two Dimension

Sales of Various Products in a City with different Quarters

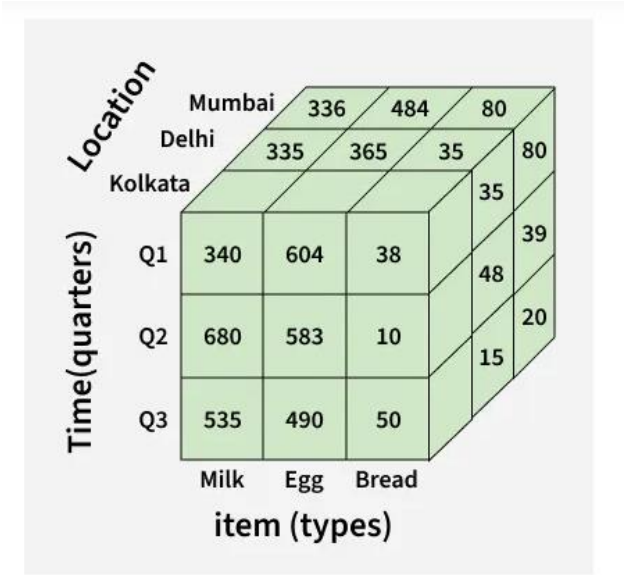
Location = "Bangalore"				
Time (quarter)	Type of item			
	Jam	Bread	Sugar	Milk
Q1	350	389	35	50
Q2	260	528	50	90
Q3	483	256	20	60
Q4	436	396	15	40

Multi Dimension

Sales of Various Products in various Cities of different quarters

	Location="Kolkata"			Location="Delhi"			Location="Mumbai"		
	item			item			item		
Time	Milk	Egg	Bread	Milk	Egg	Bread	Milk	Egg	Bread
Q1	340	604	38	335	365	35	336	484	80
Q2	680	583	10	684	490	48	595	594	39
Q3	535	490	50	389	385	15	366	385	20

The same data above, can also be represented in the form of three dimensions as given below:



3D representation of the same 2D data

Operations

Given table dimensions

- **Time:** Q1, Q2, Q3
- **Location:** Kolkata, Delhi, Mumbai
- **Item:** Milk, Egg, Bread
- **Measure:** Sales values

Roll-Up (Summarization)

Meaning: Combine data to a higher level.

Example 1: Time → Total

If we **roll-up Time**, we combine Q1, Q2, Q3.

☞ Total Milk sales in **Kolkata**:

- Q1 = 340
- Q2 = 680

- Q3 = 535
  Total = 1555

So, instead of quarter-wise data, we get **yearly total**.

Drill-Down (More Detail)

Meaning: Go from summary to detailed data.

Example

If you have **total Milk sales in Delhi**, you drill down to see:

- Q1 = 335
- Q2 = 684
- Q3 = 389

☞ Yearly → Quarter-wise details

Slice (Fix One Dimension)

Meaning: Select data for **one value of a dimension**.

Example: Slice on Time = Q1

You select only **Q1 row**.

Result:

- Kolkata → Milk 340, Egg 604, Bread 38
- Delhi → Milk 335, Egg 365, Bread 35
- Mumbai → Milk 336, Egg 484, Bread 80

Only **Q1 data**, all locations and items

Dice (Select Multiple Values)

Meaning: Select **multiple values from multiple dimensions**.

Example

Dice the cube for:

- Time: **Q1 & Q2**
- Location: **Delhi**
- Item: **Milk & Egg**

Result:

- Q1 → Milk 335, Egg 365
- Q2 → Milk 684, Egg 490

Pivot (Rotate)

Meaning: Change the way data is viewed.

Example

Current view:

- Rows → Time
- Columns → Location & Item

After **pivot**:

- Rows → Items
- Columns → Time

Data values stay same, only **orientation changes** for better analysis